

FINAL YEAR ENGINEERING PROJECT

AI-Based Real-Time Intelligent Privacy Protection System

For Live Video Streams & Broadcasts

TEAM NO.

Group-6

PROJECT GUIDE

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INTRODUCTION

The proliferation of HD surveillance cameras and live broadcasting has revolutionized security and media, but it has simultaneously created massive vulnerabilities in visual privacy.

- ▶ **Pervasive Surveillance:** Smart cities, government facilities, and corporate networks constantly record public and private spaces.
- ▶ **Live Broadcasting Risks:** News channels and military feeds often inadvertently broadcast sensitive faces, documents, or tactical locations.
- ▶ **The Manual Bottleneck:** Currently, human operators must manually scrub, delay, or censor feeds—a process that is slow, prohibitively expensive, and highly error-prone.
- ▶ **The AI Imperative:** To secure live operations, automated, low-latency Artificial Intelligence is now a strict necessity.



PROBLEM STATEMENT



Information Leaks

Sensitive military equipment, confidential boardroom meetings, classified documents, and strategic maps are routinely exposed to unauthorized viewers via unmasked camera feeds.



Identity Exposure

Protected witnesses, undercover officers, government officials, and regular civilians have their identities broadcasted globally without consent, violating strict privacy laws.



Inefficient Censorship

News broadcasters and intelligence agencies rely on delayed feeds and manual blurring. This introduces critical operational delays and inevitably allows human errors to slip through.

MOTIVATION & INSPIRATION

Military & Defense Operations

The need for armed forces to share situational awareness videos without leaking the faces of special operatives, tactical vehicle numbers, or base layouts.

Live News Broadcasting

Journalists operating in conflict zones or covering sensitive police raids require instant, automated anonymization of bystanders to comply with broadcasting ethics.

Corporate & Government Security

Protecting intellectual property on whiteboards, visible computer screens, and confidential documents during routine internal surveillance.

Global Privacy Regulations

Strict mandates like GDPR and DPDP necessitate automated methods to decouple behavioral security monitoring from biometric identification.

CORE OBJECTIVES

- ▶ **Automatic Object Detection:** Develop highly accurate deep learning models to instantly identify predefined sensitive objects in any frame.
- ▶ **Dynamic Anonymization:** Apply targeted spatial masking (blur, pixelation, blackout) strictly to the bounding box of the sensitive entity.
- ▶ **Zero Human Intervention:** Ensure the pipeline operates autonomously from frame capture to final video output.
- ▶ **Context-Aware Text Protection:** Integrate Optical Character Recognition (OCR) to detect and redact sensitive alphanumeric data (e.g., license plates, ID cards).
- ▶ **Ultra-Low Latency:** Optimize tensor operations to ensure the system processes live feeds at >30 FPS with sub-30ms latency.
- ▶ **Adaptive Tracking:** Maintain stable masking masks even when objects move rapidly, occlude each other, or change lighting conditions.
- ▶ **Scalable Architecture:** Design the software to run on local Edge AI devices (NVIDIA Jetson) as well as centralized cloud GPU servers.
- ▶ **Preserve Operational Context:** Unlike blurring the whole screen, ensure the background and non-sensitive actions remain fully visible.

THE PROPOSED SOLUTION



1. Continuous Capture

The system ingests RTSP, HTTP, or direct hardware streams from existing surveillance networks without requiring specialized cameras.



2. Intelligent AI Pipeline

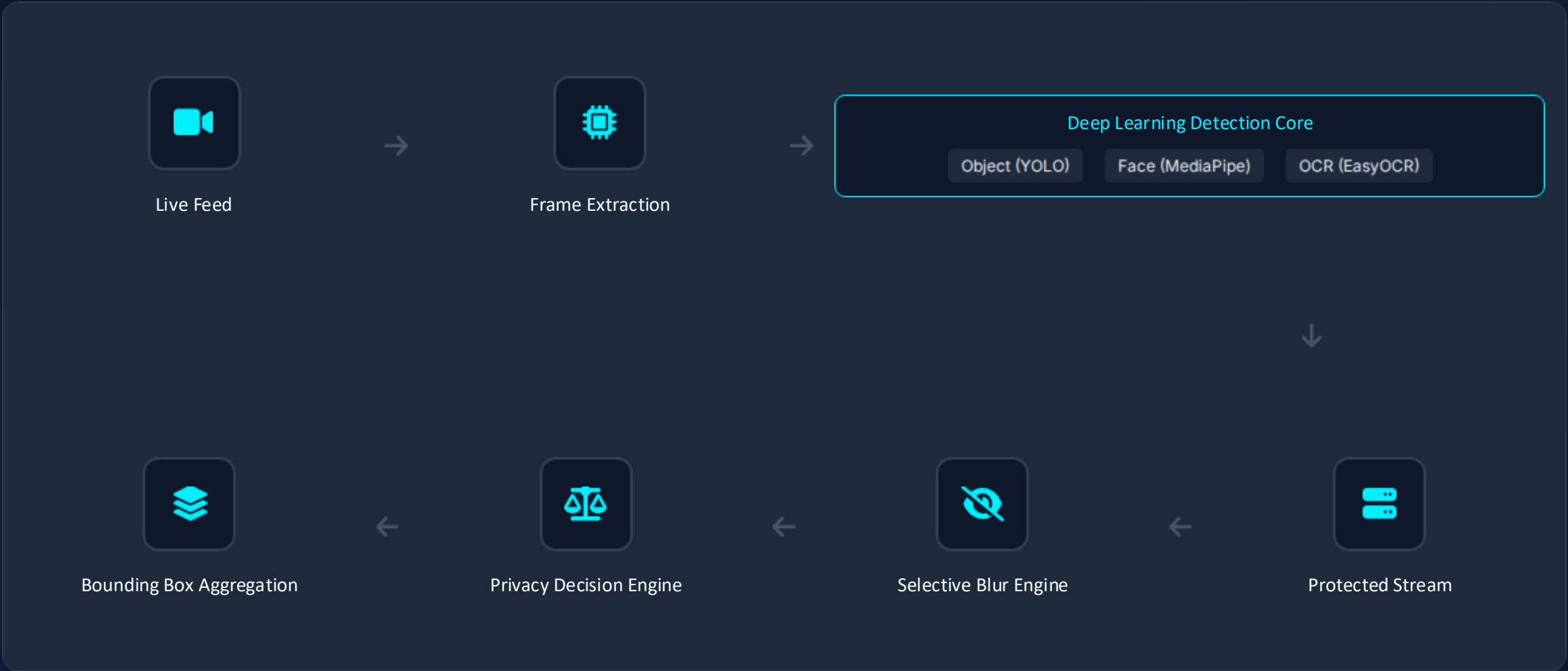
Every frame is intercepted in RAM and passed through parallelized Convolutional Neural Networks (CNNs) to localize faces, text, and classified objects simultaneously.



3. Selective Blur & Render

A decision engine determines privacy rules. Spatial masks are generated and overlaid onto the original frame instantly, yielding a secure, broadcast-ready stream.

SYSTEM ARCHITECTURE



WORKING METHODOLOGY

- ▶ **1. Preprocessing:** Incoming frames are decoded (H.264/H.265), resized, and normalized in GPU memory to match the input tensor requirements of the AI models.
- ▶ **2. Inference:** The preprocessed frame passes through optimized inference engines (TensorRT). The models output bounding box coordinates `[x, y, w, h]` and confidence scores.
- ▶ **3. Classification & Decision:** The system evaluates the detected classes (e.g., 'Face', 'Screen'). The rule engine decides if the class triggers a privacy violation based on the deployment context.
- ▶ **4. Rendering:** A Region of Interest (ROI) is extracted using the coordinates. A fast Gaussian or Pixelation kernel modifies the ROI, which is then seamlessly stitched back into the original frame.



AI & MACHINE LEARNING MODELS EVALUATED

Model Architecture	Target Domain	Speed (FPS)	Accuracy / Viability
YOLOv11 / YOLOv8	Objects, Screens, Vehicles	High (~60+ FPS)	Excellent balance of speed and AP for real-time edge deployment. Selected for core object detection.
MediaPipe Face Detection	Human Faces / Landmarks	Very High (~100 FPS)	Extremely lightweight, sub-millisecond inference. Selected for dedicated facial anonymization.
EasyOCR / PaddleOCR	Text, Documents, Plates	Moderate (~15 FPS)	High accuracy for reading confidential text, but requires batched processing to maintain live stream sync.
SAM (Segment Anything)	Pixel-perfect Masks	Low (~2-5 FPS)	Unmatched precision but computationally prohibitive for unbuffered live video processing.

DETAILED COMPUTER VISION PIPELINE



Multi-threading separates the heavy inference cycle (Steps 3-6) from the rendering cycle (Steps 7-9) to prevent stream stuttering.

SYSTEM FEATURES & CAPABILITIES

PROCESSING LATENCY

<30ms

PER FRAME OVERHEAD

- ▶ **Dynamic Adaptive Blur:** Mask intensity and size automatically scales relative to the target's distance and area within the camera's FOV.
- ▶ **Multi-Person / Multi-Object Support:** Capable of concurrently tracking and masking dozens of independent entities within a single dense frame.
- ▶ **Hardware Agnostic:** Deployable on high-end centralized cloud GPU servers or localized Edge AI chips (e.g., Jetson Nano, Coral TPU).
- ▶ **Identity & Document Protection:** Tailored specifically to intercept ID cards, faces, laptop screens, and classified tactical equipment simultaneously.

TECHNOLOGY STACK

Python



OpenCV



PyTorch



YOLO / DNN



NVIDIA CUDA



MediaPipe



EasyOCR



FastAPI



Docker



Linux OS



FFmpeg



TensorRT



TARGET APPLICATIONS

Defense & Border Security

Live command center feeds showing troop movements without exposing identities to unauthorized analysts.

News & Broadcasting

Automated face blurring of innocent bystanders and minors during live television reporting in public spaces.

Corporate Data Centers

Surveillance that monitors physical security without recording passwords or proprietary code on administrator screens.

Smart Cities & Police

Monitoring traffic flows and crowd dynamics while adhering to strict civilian biometric privacy regulations.

Healthcare

Monitoring psychiatric wards or ICU patients for physical safety without recording their identifiable faces.

Aviation & Infrastructure

Protecting the identities of security screeners and masking sensitive scanning equipment UI from public feeds.

FUTURE SCOPE

True Edge AI Integration

Moving the inference directly onto the camera hardware firmware (Smart CCTV), ensuring raw video never even touches the local network.

UAV & Drone Swarms

Applying lightweight models to military and police drone feeds, allowing live aerial reconnaissance broadcasting without revealing strategic ground assets.

Federated Learning

Training models across thousands of city cameras without centralizing video data, continuously improving detection accuracy securely.

